MySQL Operators

Introduction

MySQL operators are special symbols or keywords that perform operations on operands (values, variables, or expressions). Understanding these operators is crucial for writing effective SQL queries and performing complex database operations.

Arithmetic Operators

Arithmetic operators perform mathematical operations on numeric values.

Basic Arithmetic Operators

Operator	Description	Example
`+`	Addition	`5 + 3 = 8`
`-`	Subtraction	`10 - 4 = 6`
`*`	Multiplication	`6 * 7 = 42`
`/`	Division	`15 / 3 = 5`
`%` or `MOD`	Modulo (remainder)	`17 % 5 = 2`

Practical Examples with World Database

1. Calculate Population Density

```
-- Calculate population density (people per square km) for each country
SELECT
    Name,
    Population,
    SurfaceArea,
    ROUND(Population / SurfaceArea, 2) AS PopulationDensity
FROM country
WHERE SurfaceArea > 0
ORDER BY PopulationDensity DESC
LIMIT 10;
```

2. Calculate GNP Growth Rate

```
-- Calculate GNP growth rate percentage
SELECT
    Name,
    GNP,
    GNPOld,
    ROUND(((GNP - GNPOld) / GNPOld) * 100, 2) AS GrowthRatePercent
FROM country
```

```
WHERE GNP IS NOT NULL AND GNPOld IS NOT NULL AND GNPOld > 0
ORDER BY GrowthRatePercent DESC;
```

3. Calculate Years Since Independence

```
-- Calculate years since independence (using 2024 as current year)
SELECT
    Name,
    IndepYear,
    (2024 - IndepYear) AS YearsSinceIndependence
FROM country
WHERE IndepYear IS NOT NULL
ORDER BY YearsSinceIndependence DESC;
```

4. Population Percentage Calculations

```
-- Calculate what percentage each city represents of its country's
population
SELECT
    c.Name AS CityName,
    co.Name AS CountryName,
    c.Population AS CityPopulation,
    co.Population AS CountryPopulation,
    ROUND((c.Population / co.Population) * 100, 2) AS PercentageOfCountry
FROM city c
JOIN country co ON c.CountryCode = co.Code
WHERE co.Population > 0
ORDER BY PercentageOfCountry DESC
LIMIT 15;
```

5. Using Modulo Operator

```
-- Find countries with population that's evenly divisible by 1000000
SELECT
    Name,
    Population,
    Population % 1000000 AS Remainder
FROM country
WHERE Population % 1000000 = 0 AND Population > 0;
```

Comparison (Relational) Operators

Comparison operators compare two values and return a boolean result (TRUE/FALSE).

Comparison Operators Table

Operator	Description	Example
`=`	Equal to	`age = 25`
`!=` or `<>`	Not equal to	`status != 'active'`
`<`	Less than	`salary < 50000`

Operator	Description	Example
<=	Less than or equal	`age <= 65`
>	Greater than	`population > 1000000`
>=	Greater than or equal	`score >= 80`
<=>	NULL-safe equal	`value <=> NULL`

Practical Examples with World Database

1. Basic Comparisons

```
-- Countries with population greater than 100 million
SELECT Name, Population
FROM country
WHERE Population > 100000000
ORDER BY Population DESC;

-- Countries with exactly 0 population (data anomalies)
SELECT Name, Population
FROM country
WHERE Population = 0;

-- Countries with population not equal to their capital city population
SELECT
    co.Name AS CountryName,
    co.Population AS CountryPopulation,
    c.Name AS CapitalName,
    c.Population AS CapitalPopulation
FROM country co
JOIN city c ON co.Capital = c.ID
WHERE co.Population != c.Population;
```

2. Range Comparisons

```
-- Small countries (surface area less than 1000 sq km)
SELECT Name, SurfaceArea
FROM country
WHERE SurfaceArea < 1000 AND SurfaceArea > 0
ORDER BY SurfaceArea;

-- Countries with life expectancy 80 years or more
SELECT Name, LifeExpectancy
FROM country
WHERE LifeExpectancy >= 80
ORDER BY LifeExpectancy DESC;
```

3. NULL-Safe Comparisons

```
-- Using NULL-safe equality operator
SELECT Name, IndepYear
FROM country
WHERE IndepYear <=> NULL;

-- Compare with regular equality (shows difference)
```

```
SELECT Name, IndepYear
FROM country
WHERE IndepYear = NULL;  -- This returns no results

-- Proper NULL comparison
SELECT Name, IndepYear
FROM country
WHERE IndepYear IS NULL;
```

4. Date/Year Comparisons

```
-- Countries that gained independence after 1990
SELECT Name, IndepYear
FROM country
WHERE IndepYear > 1990
ORDER BY IndepYear;

-- Countries that gained independence before 1800
SELECT Name, IndepYear
FROM country
WHERE IndepYear < 1800
ORDER BY IndepYear;
```

Logical Operators

Logical operators combine or modify boolean expressions.

Logical Operators Table

Operator	Description	Example
`AND` or `&&`	Logical AND	`age > 18 AND status = 'active'`
`OR` or `` `		`` `
`NOT` or `!`	Logical NOT	`NOT expired`
`XOR`	Exclusive OR	`condition1 XOR condition2`

Practical Examples with World Database

1. AND Operator

```
-- Large European countries (population > 50M AND in Europe)
SELECT Name, Continent, Population
FROM country
WHERE Continent = 'Europe' AND Population > 50000000
ORDER BY Population DESC;

-- Countries with high GNP and life expectancy
SELECT Name, GNP, LifeExpectancy
FROM country
WHERE GNP > 500000 AND LifeExpectancy > 75
ORDER BY GNP DESC;
```

2. OR Operator

```
-- Countries in North America or South America
SELECT Name, Continent
FROM country
WHERE Continent = 'North America' OR Continent = 'South America'
ORDER BY Continent, Name;

-- Countries with very high or very low life expectancy
SELECT Name, LifeExpectancy
FROM country
WHERE LifeExpectancy > 80 OR LifeExpectancy < 40
ORDER BY LifeExpectancy DESC;
```

3. NOT Operator

```
-- Countries NOT in Asia
SELECT Name, Continent
FROM country
WHERE NOT Continent = 'Asia'
ORDER BY Continent, Name;

-- Countries that are NOT republics
SELECT Name, GovernmentForm
FROM country
WHERE NOT GovernmentForm LIKE '%Republic%'
ORDER BY GovernmentForm;
```

4. Complex Logical Combinations

```
-- Countries that are either very large OR very populated, but NOT in
Antarctica
SELECT Name, SurfaceArea, Population, Continent
FROM country
WHERE (SurfaceArea > 1000000 OR Population > 100000000)
  AND NOT Continent = 'Antarctica'
ORDER BY Population DESC;

-- Small island nations (small surface area AND in Oceania)
SELECT Name, SurfaceArea, Continent
FROM country
WHERE SurfaceArea < 1000 AND Continent = 'Oceania'
ORDER BY SurfaceArea;
```

String Operators

String operators work with text data.

String Operators and Functions

Operator/Function	Description	Example
`LIKE`	Pattern matching	`name LIKE 'A%'`
`NOT LIKE`	Pattern not matching	`name NOT LIKE '%land'`
`REGEXP` or `RLIKE`	Regular expression	`name REGEXP '^[A-C]'`
`CONCAT()`	String concatenation	`CONCAT(first, last)`
`LENGTH()`	String length	`LENGTH(name)`

Wildcard Characters with LIKE

Wildcard	Description	Example
`%`	Zero or more characters	`'A%'` matches "Apple", "A", "America"
`_`	Exactly one character	`'A_'` matches "AT", "AM" but not "A"

Practical Examples with World Database

1. LIKE Operator with Wildcards

```
-- Countries starting with 'A'
SELECT Name
FROM country
WHERE Name LIKE 'A%'
ORDER BY Name;

-- Countries ending with 'land'
SELECT Name
FROM country
WHERE Name LIKE '%land'
ORDER BY Name;

-- Countries containing 'stan' (many Central Asian countries)
SELECT Name
FROM country
WHERE Name LIKE '%stan%'
ORDER BY Name;

-- Countries with exactly 4 characters
SELECT Name
FROM country
WHERE Name LIKE '____'
ORDER BY Name;
```

2. NOT LIKE Operator

```
-- Countries not starting with vowels
SELECT Name
FROM country
WHERE Name NOT LIKE 'A%'
  AND Name NOT LIKE 'E%'
  AND Name NOT LIKE 'I%'
  AND Name NOT LIKE 'O%'
  AND Name NOT LIKE 'U%'
ORDER BY Name;
```

3. REGEXP (Regular Expressions)

```
-- Countries starting with A, B, or C
SELECT Name
FROM country
WHERE Name REGEXP '^[ABC]'
ORDER BY Name;

-- Countries ending with 'ia' or 'ya'
SELECT Name
```

```
FROM country
WHERE Name REGEXP '(ia|ya)$'
ORDER BY Name;

-- Countries with numbers in their name
SELECT Name
FROM country
WHERE Name REGEXP '[0-9]'
ORDER BY Name;
```

4. String Functions

```
-- Country names with their lengths
SELECT
    Name,
    LENGTH(Name) AS NameLength
FROM country
ORDER BY NameLength DESC, Name;

-- Concatenate country and continent
SELECT
    CONCAT(Name, ' (', Continent, ')') AS CountryInfo
FROM country
ORDER BY Name;

-- Cities with their country names
SELECT
    CONCAT(c.Name, ', ', co.Name) AS CityCountry
FROM city c
JOIN country co ON c.CountryCode = co.Code
ORDER BY CityCountry;
```

Assignment Operators

Assignment operators assign values to variables or columns.

Assignment Operators Table

Operator	Description	Example
`=`	Simple assignment	`SET @var = 10`
`:=`	Assignment operator	`SET @var := 10`

Practical Examples with World Database

1. Variable Assignment

```
-- Set variables and use them in queries
SET @min_population = 50000000;
SET @continent_filter = 'Asia';

SELECT Name, Population
FROM country
WHERE Population > @min_population
  AND Continent = @continent_filter
```

```
ORDER BY Population DESC;
```

2. Assignment in SELECT Statements

```
-- Running totals using assignment
SELECT
    Name,
    Population,
    @running_total := @running_total + Population AS RunningTotal
FROM country, (SELECT @running_total := 0) AS r
WHERE Continent = 'Europe'
ORDER BY Population DESC;
```

3. UPDATE with Assignment

```
-- Example of updating data (be careful with this!)
-- First, let's create a backup or work with a copy

-- Update population density in a calculated column (if it exists)
UPDATE country
SET PopulationDensity = Population / SurfaceArea
WHERE SurfaceArea > 0;
```

Special Operators

Special operators have unique functions in MySQL.

Special Operators Table

Operator	Description	Example
`BETWEEN`	Range check	`age BETWEEN 18 AND 65`
`IN`	Value in list	`city IN ('NYC', 'LA', 'Chicago')`
`IS NULL`	NULL check	`phone IS NULL`
`IS NOT NULL`	NOT NULL check	`email IS NOT NULL`
`EXISTS`	Subquery exists	`EXISTS (SELECT...)`
`CASE`	Conditional logic	`CASE WHEN...THEN...END`

Practical Examples with World Database

1. BETWEEN Operator

```
-- Countries with medium population (between 10M and 50M)
SELECT Name, Population
FROM country
WHERE Population BETWEEN 10000000 AND 50000000
ORDER BY Population DESC;

-- Countries that gained independence between 1950 and 1990
SELECT Name, IndepYear
FROM country
WHERE IndepYear BETWEEN 1950 AND 1990
```

```
ORDER BY IndepYear;
```

2. IN Operator

```
-- Countries in specific continents
SELECT Name, Continent
FROM country
WHERE Continent IN ('Europe', 'Asia', 'North America')
ORDER BY Continent, Name;

-- Major languages
SELECT CountryCode, Language, Percentage
FROM countrylanguage
WHERE Language IN ('English', 'Spanish', 'French', 'German', 'Chinese')
ORDER BY Language, Percentage DESC;
```

3. IS NULL and IS NOT NULL

```
-- Countries with unknown independence year
SELECT Name, IndepYear
FROM country
WHERE IndepYear IS NULL
ORDER BY Name;

-- Countries with known life expectancy
SELECT Name, LifeExpectancy
FROM country
WHERE LifeExpectancy IS NOT NULL
ORDER BY LifeExpectancy DESC;
```

4. EXISTS Operator

```
-- Countries that have cities in the database
SELECT Name
FROM country co
WHERE EXISTS (
    SELECT 1 FROM city c WHERE c.CountryCode = co.Code
)
ORDER BY Name;

-- Countries with no cities recorded
SELECT Name
FROM country co
WHERE NOT EXISTS (
    SELECT 1 FROM city c WHERE c.CountryCode = co.Code
)
ORDER BY Name;
```

5. CASE Statement

```
-- Categorize countries by population size
SELECT
    Name,
    Population,
    CASE
        WHEN Population > 100000000 THEN 'Very Large'
        WHEN Population > 50000000 THEN 'Large'
        WHEN Population > 10000000 THEN 'Medium'
```

```
        WHEN Population > 1000000 THEN 'Small'
        ELSE 'Very Small'
    END AS PopulationCategory
FROM country
WHERE Population > 0
ORDER BY Population DESC;

-- Categorize countries by continent regions
SELECT
    Name,
    Continent,
    CASE Continent
        WHEN 'Asia' THEN 'Eastern Hemisphere'
        WHEN 'Europe' THEN 'Western Civilization'
        WHEN 'Africa' THEN 'African Continent'
        WHEN 'North America' THEN 'Americas'
        WHEN 'South America' THEN 'Americas'
        WHEN 'Oceania' THEN 'Pacific Region'
        ELSE 'Other'
    END AS RegionCategory
FROM country
ORDER BY RegionCategory, Name;
```

Operator Precedence

Understanding operator precedence is crucial for writing correct SQL expressions.

Precedence Order (Highest to Lowest)

1. **Parentheses** `()`
2. **Unary operators** `+`, `-`, `~`, `!`
3. **Multiplicative** `*`, `/`, `%`, `MOD`
4. **Additive** `+`, `-`
5. **Shift operators** `<<`, `>>`
6. **Bitwise AND** `&`
7. **Bitwise XOR** `^`
8. **Bitwise OR** `|`
9. **Comparison** `=`, `!=`, `<`, `<=`, `>`, `>=`, `<=>`, `LIKE`, `IN`, `BETWEEN`
10. **Logical NOT** `NOT`
11. **Logical AND** `AND`, `&&`
12. **Logical XOR** `XOR`
13. **Logical OR** `OR`, `||`
14. **Assignment** `=`, `:=`

Practical Examples

```
-- Without parentheses (follows precedence)
SELECT Name, Population
FROM country
WHERE Population > 10000000 AND Continent = 'Asia' OR Continent = 'Europe';
-- This is evaluated as: (Population > 10000000 AND Continent = 'Asia') OR
Continent = 'Europe'
```

```
-- With parentheses (explicit grouping)
SELECT Name, Population
FROM country
WHERE Population > 10000000 AND (Continent = 'Asia' OR Continent =
'Europe');
-- This gives different results!

-- Mathematical operations precedence
SELECT
    Name,
    Population,
    SurfaceArea,
    Population / SurfaceArea * 100 AS DensityPercent  -- Division first,
then multiplication
FROM country
WHERE SurfaceArea > 0
LIMIT 10;
```